Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 8

Question 4

```
Delete x from T;
n ← n − 1;
If (n = 0)
        free(T);
Else
     If(n = size(T)/2)
   {    T′ ← createTable( n/2);
        copy(T,T′);
        free(T);
        T ← T′
   }
```

Note, T is the dynamic table that supports only deletions.

Using Potential method show that the amortized cost of each Deletion operation is O(1).

(State your potential function.)